DR. RISHI GURNANI

U.S. Citizen | rishipgurnani@gmail.com | rishigurnani.github.io | rishigurnani | >2,100 citations

SUMMARY

Brief: 10+ years experience in computer programming and data science; 10+ years experience in scientific computing (drug discovery, chemistry, materials science, physics); 8+ years experience in deep learning; 8+ years experience in research; 2+ years experience as a people leader

Long: Data science expert specializing in generative modeling for small molecule and materials discovery. Ph.D. from Georgia Tech; 16 publications (>2,100 citations, h-index 16), including first-author papers in *Nature Communications* and *Chemistry of Materials*. Proven ability to communicate complex AI methods to both technical and non-technical audiences — as a people leader, keynote speaker, podcast guest, and invited lecturer at universities and national labs. Deep familiarity with the academic and research ecosystem in AI for scientific computing.

CORE COMPETENCIES

Generative AI	VAE, GAN, graph-based generative models; applied to molecular and polymer design
Deep Learning	GNNs, multitask learning, physics-informed ML, model ensembling, inference optimization
Statistical	Regression, classification, uncertainty quantification, feature engineering, hypothesis testing, statistical modeling
Research Ecosystem	Active knowledge of leading labs, researchers, and open problems in AI for chemistry & materials
Programming	Python, PyTorch, PyTorch Geometric, RDKit, NumPy, Scikit-learn, Pandas, FastAPI, HuggingFace, PyTest
Open Source	`polygnn_trainer` (PyTorch ML training library, adopted by materials informatics community)

WORK EXPERIENCE

Senior Manager, Data Science for Global Ingredient Research — Coca-Cola Feb 2025 – present
Atlanta, GA

- Lead AI strategy and core algorithm development for global ingredient R&D (sweeteners, flavors, emulsifiers, wellness ingredients), spanning small molecules to formulations
- Promise Award Finalist (2026): Developed and trained the AI models behind AIRE 3.0, a beverage-design engine that combines flavorist sensory knowledge with physics-based color prediction, cutting limited-time-offer (LTO) beverage creation from 3 months to 30 min; extended the models to design drinks that pair with specific restaurant menus, acheiving 93% taste-test approval (+18% above target)
- Built an in-house cell assay data pipeline from scratch, harmonizing 10+ years of legacy assay data across major protocol changes into a unified, ML-ready dataset that was previously unusable for modeling
- Apply LLMs, sentence transformers, PLS regression, and recommender systems to cell assay, clinical trial, docking, and sensory data (OpenAI, PyTorch, scikit-learn, FastAPI, HuggingFace, RDKit)

Director of Software Engineering & Algorithms — Matmerize Inc. Mar 2021 – Feb 2025
Atlanta, GA

- Developed the deep learning core of PolymRize™, an enterprise ML platform (∼100K lines of code, 5 languages) adopted by >20 research organizations and companies.
- Served as primary technical point of contact for academic and industry partners, translating research needs into software capabilities and communicating algorithmic methods to audiences ranging from wet-lab chemists to ML engineers.
- Managed junior developers and interns; alumni now at **Meta**, **UC Berkeley (PhD)**, and **Georgia Tech (PhD)**.

EDUCATION

Ph.D., Materials Science & Engineering — Georgia Tech (GPA 4.00)

Dissertation: Advances in AI for Accelerated Discovery of Energy Storage Polymers (GNNs & Generative ML)

M.S., Computer Science — Georgia Tech (Specialization: Machine Learning)

B.S., Materials Science & Engineering — UC Berkeley

AWARDS, TALKS & OPEN SOURCE

Best Ph.D. Thesis, Georgia Tech Sigma Xi

Graduate Student Award, Materials Research Society

Gold Medal, Best Presentation, 2nd Energy & Informatics International Forum

Invited Talks: Keynote, 2024 RISE Conference; Tokyo Institute of Technology (2022); Sandia National Laboratory (2021); nanoHUB ML Series (2021); STLE 76th Annual Meeting (2022). Podcast: *It's a Material World* (2023).

Open Source: `polygnn_trainer` — PyTorch library for automated ML training, adopted by the materials informatics community. `fintracker` — open-source personal finance analytics library. `rock-ur-mock` — Mock draft simulator for competitive fantasy football players.

PUBLICATIONS (16 PAPERS, >2,100 CITATIONS, H-INDEX 16)

1. AI-assisted discovery of high-temperature dielectrics for energy storage. R Gurnani, S Shukla et al. *Nature Communications* 15 (1), 2024. *(93 citations)*
2. Polymer informatics at scale with multitask graph neural networks. R Gurnani, C Kuenneth, A Toland, R Ramprasad. *Chemistry of Materials* 35 (4), 2023. *(106 citations)*
3. PolyG2G: A novel machine learning algorithm applied to the generative design of polymer dielectrics. R Gurnani, D Kamal, H Tran et al. *Chemistry of Materials* 33 (17), 2021. *(82 citations)*
4. Design of functional and sustainable polymers assisted by artificial intelligence. H Tran, R Gurnani, C Kim et al. *Nature Reviews Materials* 9 (12), 2024. *(198 citations)*
5. Stereoisomerism of Vicinal Polydichloronorbornene for Ultra-High-Temperature Capacitive Energy Storage. J Hao, S Shukla, R Gurnani et al. *Advanced Materials* 37 (15), 2025.
6. A physics-enforced neural network to predict polymer melt viscosity. A Jain, R Gurnani, A Rajan, HJ Qi, R Ramprasad. *npj Computational Materials* 11 (1), 2025.
7. Rationally designed high-temperature polymer dielectrics for capacitive energy storage: An experimental and computational alliance. PS Aklujkar, R Gurnani et al. *Progress in Polymer Science* 161, 2025.
8. Polymer composites informatics for flammability, thermal, mechanical and electrical property predictions. H Tran, C Kim, R Gurnani et al. *Polymer Chemistry* 16 (30), 2025.
9. Gas permeability, diffusivity, and solubility in polymers: Simulation-experiment data fusion and multi-task machine learning. BK Phan, KH Shen, R Gurnani et al. *npj Computational Materials* 10 (1), 2024.
10. Pendant group functionalization of cyclic olefin for high temperature and high-density energy storage. S Shukla, C Wu, A Mishra, R Gurnani et al. *Advanced Materials* 36 (52), 2024.
11. Effect of fluorine in redesigning energy-storage properties of high-temperature dielectric polymers. AA Deshmukh, C Wu, R Gurnani et al. *ACS Applied Materials & Interfaces* 15 (40), 2023.
12. Identifying High-Performance Metal-Organic Frameworks for Low-Temperature Oxygen Recovery from Helium by Computational Screening. S Jamdade, R Gurnani, H Fang, SE Boulfelfel, R Ramprasad, DS Sholl. *Industrial & Engineering Chemistry Research* 62 (4), 2023.
13. Interpretable machine learning-based predictions of methane uptake isotherms in metal-organic frameworks. R Gurnani, Z Yu, C Kim, DS Sholl, R Ramprasad. *Chemistry of Materials* 33 (10), 2021.
14. Machine-learning predictions of polymer properties with Polymer Genome. H Doan Tran, C Kim, L Chen, R Gurnani et al. *Journal of Applied Physics* 128 (17), 2020. *(320 citations)*
15. Experimental characterization of the shear properties of 3D-printed ABS and polycarbonate parts. S Rohde, J

Cantrell, R Gurnani et al. *Experimental Mechanics* 58 (6), 2018.

16. Experimental characterization of the mechanical properties of 3D-printed ABS and polycarbonate parts. JT Cantrell, S Rohde, D Damiani, R Gurnani et al. *Rapid Prototyping Journal* 23 (4), 2017. *(777 citations)*